D06 · Cold Chain & Specialized Logistics

# 05 · E-commerce fulfillment logistics & last mile

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

## 1 WHAT IT IS

E-commerce fulfillment and last mile cover how online orders are picked, packed and delivered to homes, lockers or pickup points — from own DCs, 3PLs, marketplaces or stores — at a cost and speed the customer accepts.

## 2 WHY IT MATTERS

Last mile is usually the most expensive leg per unit because it delivers small parcels to dispersed addresses with failed attempts and returns. **Its economics are driven by density: drops per kilometer and first-attempt success.** In Mexican cities, access restrictions, security and incomplete addresses make out-of-home delivery and address validation unusually valuable.

## 3 HOW IT IS DONE

1. **Measure cost per delivered order by zone.** Include failed attempts and returns, not only the carrier rate.
2. **Raise density deliberately.** Delivery windows, lockers and pickup points, and order batching by zone.
3. **Validate addresses and communicate.** Clean data at checkout and proactive notifications cut failed attempts.
4. **Allocate orders across nodes dynamically.** Choose DC, 3PL or store by cost and promise for each order.

### IN PRACTICE

**Treat density as a first-class design variable**

Every initiative that increases drops per km or reduces failed attempts improves the fundamental economics.

**Design for hybrid fulfillment from the start**

Pure single-node models rarely remain optimal as volume and geographic coverage grow.

**Integrate returns into the same network**

Out-of-home points that also accept returns improve both customer experience and reverse-logistics cost.

### ILLUSTRATIVE CASE

*Illustrative case built from typical industry values — not data from a specific company.*

**Illustrative case — Mid-sized e-commerce player in Mexico** — Pure home-delivery model with rising last-mile cost and 18% failed-delivery rate on first attempt.

| Initiative | Impact |
|---|---|
| Introduction of locker + collection-point network in top 8 cities | First-attempt success +11 pp · Last-mile cost -14% on covered volume |
| Dynamic allocation between own DC and 3PL nodes | Average delivery distance -18% |
| Improved address validation + customer communication | Failed deliveries -23% |

**Result:** Blended last-mile cost per order reduced 17% while average delivery speed improved. The locker network also reduced reverse-logistics cost because returns could be dropped at the same points.

### HOW IT IS MEASURED

**Last-Mile Cost per Order**

Total last-mile cost / Number of delivered orders

*Benchmark: Highly dependent on density and model; track trend and compare across nodes*

**First-Attempt Delivery Success %**

(Orders delivered on first attempt / Total delivery attempts) × 100

*Benchmark: >90% is a solid target in urban environments with good address data*

#### WHAT YOU WOULD USE

**E-commerce & Last-Mile Platforms**

**Delivery platforms + locker networks (local and regional providers)** — Critical for out-of-home density in Mexican cities.

**OMS + TMS with multi-node allocation** — Order management systems that can intelligently choose fulfillment node and last-mile method.

## 4 SO WHAT

> **Win the last mile on density and first-attempt success.** Measure cost per delivered order, add out-of-home options, fix addresses at checkout and route each order to the right node.

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